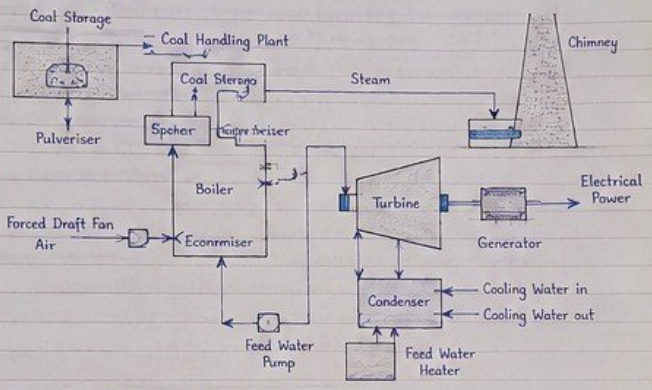
1. Explain the layout diagram of Thermal Power Plant and Describe each part in detail.

Layout Diagram :-

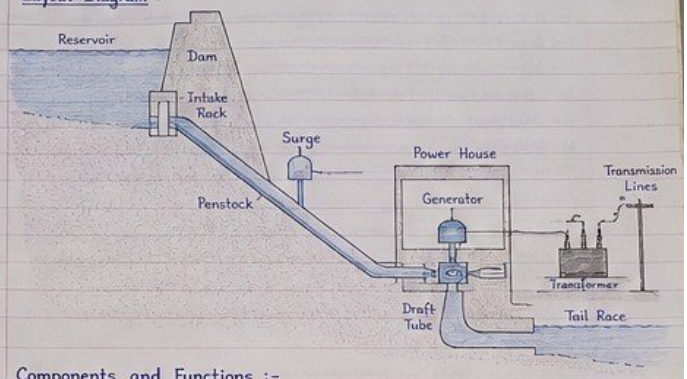


Main Components and Functions :-

1. **Coal Handling Plant** :- Stores and supplies coal to the boiler.
2. **Boiler** :- Converts water into steam using heat produced by burning coal.
3. **Superheater** :- Superheats the steam to high temperature.
4. **Turbine** :- Converts heat energy of steam into mechanical energy.
5. **Generator** :- Converts mechanical energy into electrical energy.
6. **Condenser** :- Converts exhaust steam back into water.
7. **Feed Water Pump** :- Pumps water from condenser to boiler.
8. **Economiser/Feed Water Heater** :- Heats feed water to improve efficiency.
9. **Cooling Tower** :- Cools the hot water used in condenser.
10. **Chimney** :- Discharges harmful flue gases to atmosphere.

2. Draw the Hydro electric power plant layout and Explain all its components.

Layout Diagram :-

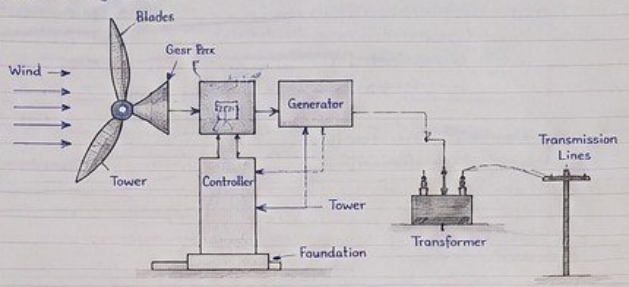


Components and Functions :-

1. **Dam** :- Stores water and creates head.
2. **Reservoir** :- Stores large quantity of water.
3. **Intake & Trash Rack** :- Allows water to enter and removes debris.
4. **Penstock** :- Carries water to turbine under pressure.
5. **Surge Tank** :- Prevents water hammer and pressure fluctuations.
6. **Turbine** :- Converts kinetic energy of water into mechanical energy.
7. **Generator** :- Converts mechanical energy into electrical energy.
8. **Draft Tube** :- Recovers energy from water and directs it to tail race.
9. **Tail Race** :- Returns used water back to the river.
10. **Transformer** :- Steps up voltage for transmission.
11. **Transmission Lines** :- Carry electricity to consumers.

3. Draw the layout diagram of wind power plant and Explain each part.

Layout Diagram :-

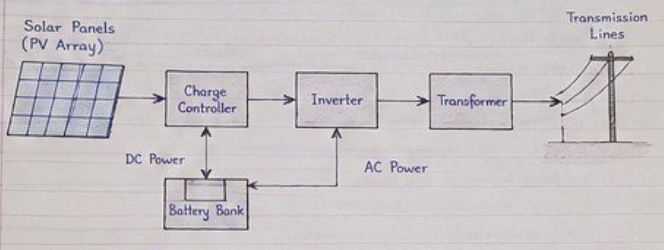


Components and Functions :-

1. **Blades** :- Capture wind energy and rotate.
2. **Rotor** :- Supports blades and transmits rotation.
3. **Gear Box** :- Increases speed from rotor to generator.
4. **Generator** :- Converts mechanical energy into electrical energy.
5. **Controller** :- Controls starting, stopping and safe operation.
6. **Tower** :- Supports nacelle and rotor at height.
7. **Transformer** :- Steps up voltage for transmission.
8. **Foundation** :- Provides stability to the tower.
9. **Transmission Lines** :- Carry power to consumers.

4. Draw the layout diagram of Solar power plant and Explain each and every component.

Layout Diagram :-



Components and Functions :-

1. **Solar Panels** :- Convert sunlight into D. C electrical energy.
2. **Charge Controller** :- Regulates voltage and current to battery.
3. **Battery Bank** :- Stores energy for future use.
4. **Inverter** :- Converts DC power into AC power.
5. **Transformer** :- Steps up voltage for transmission.
6. **Transmission Lines** :- Carry power to consumers.

5. What is Electric shock. Explain the safety prication to avoid electric shock.

Electric Shock :- It is the effect produced on human body when electric current passes through it. It may cause injury, burns, muscle contraction or even death.

Safety Precautions :-

1. Proper earthing of electrical equipment.
2. Use of good quality insulated wires.
3. Do not touch electrical appliances with wet hands.
4. Use of protective devices like fuse, MCB, ELCB.
5. Switch off power supply before repairing.
6. Use of rubber gloves, shoes and insulated tools.
7. Keep electrical appliances away from water.
8. Do not overload electrical circuits.
9. Regular inspection and maintenance.
10. Keep children away from electrical equipment.

First Aid :-

1. Switch off the power supply.
2. Do not touch the person directly.
3. Move the person away using dry stick or wood.
4. Give artificial respiration if needed and call doctor.

6. Explain the operation of Fuse and MCB.

(A) Fuse :-

Operation :-

- Fuse is connected in series.
- When excessive current flows, fuse element gets heated.
- It melts and breaks the circuit.
- Thus the supply is disconnected and appliances are protected.

Advantages :-

1. Simple and cheap.
2. Easy to replace.

Disadvantages :-

1. Has to be replaced after operation.
2. Cannot be reused.

(B) MCB (Miniature Circuit Breaker) :-

Operation :-

- MCB works on thermal and magnetic effects.
- In overload, bimetallic strip heats and bends to trip the circuit.
- In short circuit, electromagnetic coil attracts and opens the contacts.
- Thus circuit is opened and supply is disconnected.

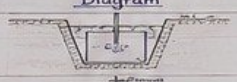
Advantages :-

1. Reusable.
2. Automatic operation.
3. More reliable and safe.

Disadvantages :-

1. Costlier than fuse.
2. Slightly complex.

7. Explain different types of Earthing.

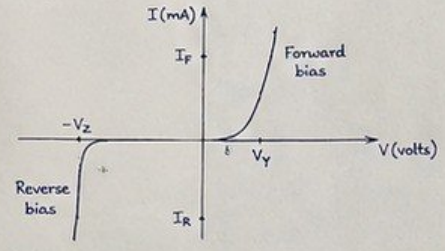
Type of Earthing	Diagram	Construction	Advantages	Disadvantages	Applications
1. Plate Earthing		A copper/GI plate is buried vertically in the ground with charcoal and salt.	• Low resistance • More reliable	• Costly • Needs more space	• Substations, industries
2. Pipe Earthing		A perforated GI pipe is buried vertically with charcoal and salt.	• Economical • Simple to install	• Corrosion of pipe	• Buildings, small installations
3. Rod Earthing		A copper/GI rod is driven into the ground.	• Easy installation • Suitable for rocky areas	• Higher resistance in dry soil	• Transmission lines, poles
4. Strip Earthing		A copper/GI strip is buried horizontally in the ground.	• Good for large installations	• Requires more area	• Large installations, factories
5. Chemical Earthing		Electrode is surrounded by chemical compound to reduce resistance.	• Very low resistance • Low maintenance	• High initial cost	• Modern power systems

1. Explain the V-I characteristics of a PN junction diode. Discuss forward and reverse bias, breakdown mechanisms.

Ans:

V-I Characteristics of a PN Junction Diode:

The V-I characteristic curve shows the relation between voltage across the diode and current through it.



Forward Bias:

- P-side is connected to +ve terminal and N-side to -ve terminal.
- Depletion width decreases and barrier potential reduces.
- Current starts flowing after cut-in voltage (knee voltage).
- Silicon diode : $V_\gamma \approx 0.7$ V
- Germanium diode : $V_\gamma \approx 0.3$ V

Reverse Bias:

- P-side is connected to -ve terminal and N-side to +ve terminal.
- Depletion width increases and barrier potential increases.
- Only a small reverse saturation current (I_s) flows due to minority carriers.
- If reverse voltage exceeds a certain value, breakdown occurs.

Breakdown Mechanisms:

(a) Zener Breakdown : Heavily doped diode, thin depletion region, occurs at low reverse voltage due to strong electric field.

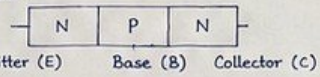
(b) Avalanche Breakdown : Lightly doped diode, wide depletion region, occurs at high reverse voltage due to collision of charge carriers.

2. With the schematic and symbolic diagram of NNN and PNP transistor and describe their operation.

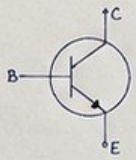
Ans:

(a) NPN Transistor

Structure:



Symbol:



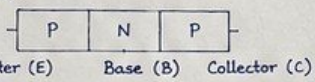
Operation:

- Emitter-base junction is forward biased.
- Collector-base junction is reverse biased.
- Electrons are emitted from emitter, most of them reach collector.
- Small base current controls large collector current.

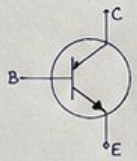
$$I_E = I_B + I_C$$

(b) PNP Transistor

Structure:



Symbol:



Operation:

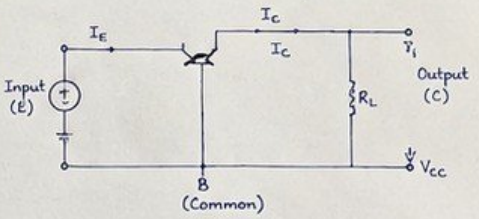
- Emitter-base junction is forward biased.
- Collector-base junction is reverse biased.
- Holes are emitted from emitter, most of them reach collector.
- Small base current controls large collector current.

$$I_E = I_B + I_C$$

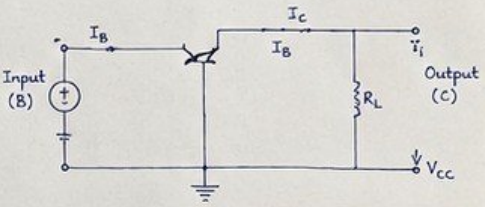
3. Draw the schematic of CB, CC, CE configurations of BJT and analyse their performances.

Ans:

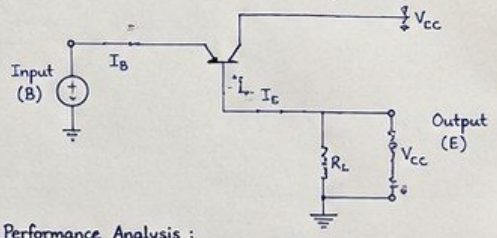
(a) Common Base (CB) Configuration



(b) Common Emitter (CE) Configuration



(c) Common Collector (CC) Configuration



Performance Analysis:

Parameter	CB	CE	CC
Input Resistance	Low	Medium	High
Output Resistance	High	Medium	Low
Current Gain (A_i)	< 1	High	High
Voltage Gain (A_v)	High	High	≈ 1
Power Gain	Medium	Very High	Medium
Phase Shift	0°	180°	0°

4. Explain the operation of Zener diode and draw V-I characteristics.

Ans:

Construction:

- Heavily doped PN junction diode.
- Very thin depletion region.
- Designed to operate in reverse breakdown region.

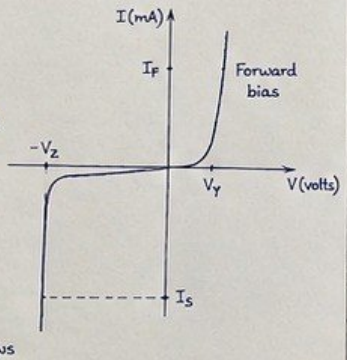
Operation:

(i) Forward Bias:

- Acts like normal diode.
- Current flows after cut-in voltage (≈ 0.7 V for Si, 0.3 V for Ge).

(ii) Reverse Bias:

- Only small reverse saturation current flows initially.
- When reverse voltage reaches Zener voltage (V_Z), breakdown occurs.
- Current increases sharply while voltage remains almost constant.
- This property is used for voltage regulation.



Breakdown Mechanisms:

(a) Zener Breakdown : Heavily doped, low reverse voltage, due to strong electric field.

(b) Avalanche Breakdown : Lightly doped, high reverse voltage, due to collision of carriers.

Applicateristics:

1. Voltage regulation
2. Over-voltage protection
3. Voltage reference circuits
4. Wave shaping circuits
5. Surge suppressors

5. Compare diode with Zener diode and write the applications.

Ans:

Comparison Table:

Feature	PN Junction Diode	Zener Diode
Definition	Ordinary semiconductor diode	Special diode designed for reverse breakdown
Doping	Moderately doped	Heavily doped
Depletion Region	Wider	Very thin
Operation	Mainly in forward bias	Mainly in reverse bias
Reverse Breakdown	May damage the diode	Safe and controlled
Breakdown Voltage	High	Low and fixed
Current Flow	One direction only	Reverse current after breakdown
Symbol		
Main Function	Rectification	Voltage regulation

Applications:

(a) PN Junction Diode:

1. Rectifiers in power supplies
2. Clippers and clampers
3. Signal detection
4. Switching circuits
5. Demodulation circuits

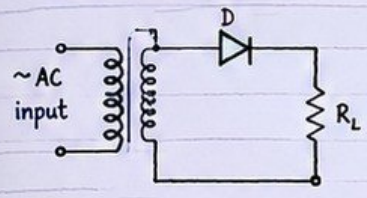
(b) Zener Diode:

1. Voltage regulation
2. Voltage reference circuits
3. Over-voltage protection
4. Surge suppressors
5. Wave shaping circuits

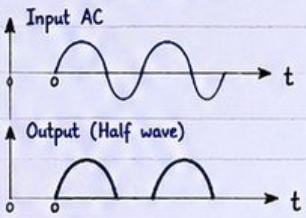
Electronic Devices and Circuits – University Answers

1. Describe half wave and full wave rectifiers. Explain their operation with neat waveforms.

a) Half wave rectifier

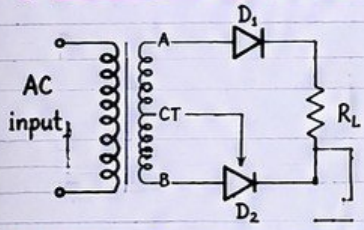


Waveforms

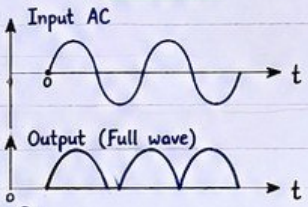


- **Operation :**
  - During +ve half cycle diode is forward biased and current flows through R_L.
  - During -ve half cycle diode is reverse biased and no current flows.

b) Full wave rectifier (Centre tapped)



Waveforms



- **Operation :** D_1 conducts in +ve half cycle, D_2 conducts in -ve half cycle. Current through R_L is in the same direction during both cycles.

2. Derive expressions for the DC output voltage of half wave and full wave and bridge rectifiers and compare their performance characteristics.

(a) Half wave rectifier

$$V_{DC} = \frac{1}{2\pi}\int_0^{\pi} V_m \sin\theta \, d\theta = \frac{V_m}{\pi} = 0.318 V_m$$

(b) Full wave rectifier

$$V_{DC} = \frac{1}{\pi}\int_0^{\pi} V_m \sin\theta \, d\theta = \frac{2V_m}{\pi} = 0.636 V_m$$

(c) Bridge rectifier

$$V_{DC} = \frac{2V_m}{\pi} = 0.636 V_m$$

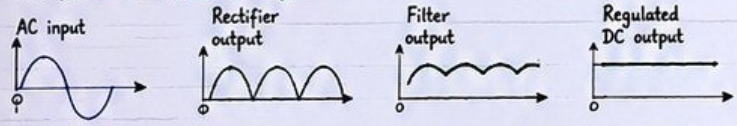
Comparison Table

Parameter	Half wave	Full wave (Centre tapped)	Bridge rectifier
Diodes used	1	2	4
V_{DC} (avg.)	V_m/π ($0.318V_m$)	$2V_m/\pi$ ($0.636V_m$)	V_m/π ($0.636V_m$)
Efficiency (η)	40.6%	81.2%	81.2%
Ripple factor (r)	1.21	0.482	0.482
PIV per diode	V_m	$2V_m$	V_m
Output freq.	f	2f	2f
Transformer utilization	Low	Better	Highest

3. Draw and explain the block diagram of a DC power supply.



Waveforms at different stages :



Functions :

1. Transformer – changes voltage level and provides isolation.
2. Rectifier – converts AC to pulsating DC.
3. Filter – removes ripples and gives smooth DC.
4. Regulator – maintains constant DC output.

4. Explain the gray and hamming codes with an example.

(a) Gray Code

Only one bit changes between two consecutive numbers.

Example : Binary 1011

Gray code conversion

1. First bit same = 1
2. Next bits = XOR of adjacent bits

1011 (Binary)

1 ⊕0=1 ⊕1=1 ⊕1=0

Gray Code = 1110

Gray Code Table

Decimal	Binary	Gray
0	000	000
1	001	001
2	010	011
3	011	010
4	100	110
5	101	111
6	110	101
7	111	100

(b) Hamming Code (Even Parity)

Data bits = 1011, m=4, parity bits = 3 (positions 1,2,4)

Arrangement :

Position :	1	2	3	4	5	6	7
Bit :	P_1	P_2	1	P_4	0	1	1

Parity calculation :

- P_1 (1,3,5,7) : 1 0 1 → even → $P_1 = 0$
- P_2 (2,3,6,7) : 1 1 1 → odd → $P_2 = 1$
- P_4 (4,5,6,7) : 0 1 1 → even → $P_4 = 0$

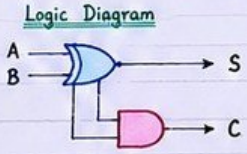
Hamming code = 0110011

5. Write a short notes on half adder and full adder.

Half Adder

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Logic Diagram

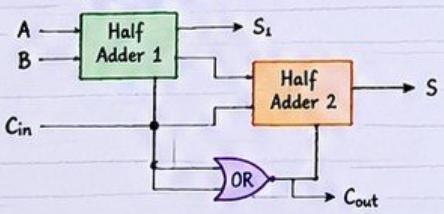


Equations : $S = A \oplus B$, $C = A \cdot B$

Full Adder

A	B	Cin	S	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Logic Diagram



Equations : $S = A \oplus B \oplus C_{in}$

$C_{out} = AB + AC_{in} + BC_{in}$

6. Explain in detail all basic logic gates : AND, OR, NOT. Draw the logic symbol, truth table.

AND Gate



$Y = A \cdot B$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

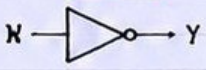
OR Gate



$Y = A + B$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

NOT Gate



$Y = \overline{A}$

A	Y
0	1
1	0

Summary : AND gives 1 only when all inputs are 1.
OR gives 1 when any one input is 1.
NOT gives complement of input.